



Cataloging to Support Information Literacy: The IFLA Library Reference Model's User Tasks in the Context of the *Framework for Information Literacy for Higher Education*

C. Rockelle Strader

To cite this article: C. Rockelle Strader (2021) Cataloging to Support Information Literacy: The IFLA Library Reference Model's User Tasks in the Context of the *Framework for Information Literacy for Higher Education*, *Cataloging & Classification Quarterly*, 59:5, 442-476, DOI: [10.1080/01639374.2021.1939828](https://doi.org/10.1080/01639374.2021.1939828)

To link to this article: <https://doi.org/10.1080/01639374.2021.1939828>



© 2021 The Author(s). Published with license by Taylor & Francis Group, LLC.



Published online: 30 Jun 2021.



Submit your article to this journal [↗](#)



Article views: 5492



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 3 View citing articles [↗](#)

Cataloging to Support Information Literacy: The IFLA Library Reference Model's User Tasks in the Context of the *Framework for Information Literacy for Higher Education*

C. Rockelle Strader 

University Libraries, The Ohio State University, Columbus, Ohio, USA

ABSTRACT

Cataloging practices, as exemplified by the five user tasks of the IFLA Library Reference Model, can support information literacy practices. The six frames of the *Framework for Information Literacy for Higher Education* are used as lenses to examine the user tasks. Two themes emerge from this examination: context matters, and catalogers must tailor bibliographic descriptions to meet users' expectations and information needs. Catalogers need to solicit feedback from various user communities to reform cataloging practices to remain current and viable. Such conversations will enrich the catalog and enhance (reclaim?) its position as a primary tool for research and learning.

KEYWORDS

Cataloging;
Framework for Information Literacy for Higher Education;
information literacy;
IFLA Library Reference Model; LRM; user tasks; users

Introduction

In 2016 the Association of College and Research Libraries (ACRL) approved and released the *Framework for Information Literacy for Higher Education* for use across higher education information and teaching environments. Its purpose was to

reflect the current thinking on such things as the creation and dissemination of knowledge, the changing global higher education and learning environment, the shift from information literacy to information fluency, and the expanding definition of information literacy to include multiple literacies, for example, transliteracy, media literacy, digital literacy, etc.¹

A formal definition of information literacy was presented in the *Framework*, as follows:

Information literacy is the set of integrated abilities encompassing the reflective discovery of information, the understanding of how information is produced and

CONTACT C. Rockelle Strader  strader.2@osu.edu  The Ohio State University Libraries, 1165 Kinnear Rd., Columbus, OH 43212, USA.

© 2021 The Author(s). Published with license by Taylor & Francis Group, LLC.

This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (<http://creativecommons.org/licenses/by-nc-nd/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way.

valued, and the use of information in creating new knowledge and participating ethically in communities of learning.²

A year later, in 2017, the International Federation of Library Associations and Institutions (IFLA) approved and released the IFLA Library Reference Model (LRM) to the cataloging community. It was designed to consolidate three previous entity-relationship models (discussed below). The intended scope of the model is to be user-facing, so the “user tasks” associated with each of the three previous models were consolidated as well. As noted in the LRM’s discussion about users and user tasks, “The model is primarily concerned with the data and functionality required by end-users (and intermediaries working on behalf of end-users) to meet their information needs.”³

Also in 2017, at ACRL’s conference, Erin Leach presented “Using the *Framework* to Frame: Cataloging Policy and Practice as Seen through the Lens of *The Framework for Information Literacy for Higher Education*.” In her talk, she used the *Framework* as a rubric “to view goals and currently accepted practices of cataloging.”⁴ Her critique brings forward an uncomfortable view that cataloging tends to exist in a silo that interoperates inconsistently with public-facing services and goals, such as those represented by the *Framework for Information Literacy for Higher Education*.

Leach’s presentation was a critique of cataloging policy and practice in general using the *Framework* as context. This article is intended to focus more specifically on the LRM user tasks in the context of the ACRL *Framework*. The LRM user tasks will be placed in each of the six “frames” of the *Framework* to suggest how cataloging processes may facilitate information literacy. The aim is to harmonize activities of cataloging with precepts of information literacy and to discuss issues about providing consistent access to resources for meeting the information needs of our users. The LRM user tasks have the potential to act as guides for supporting information literacy via the catalog.

In this article, the LRM user tasks will be introduced first, with brief general comments on each task. Then the *Framework* will be introduced, with comments on cataloging issues related to each frame. Finally, each user task, as informed by the context of each frame, will be presented.

Library Reference Model (LRM) user tasks

The Library Reference Model (LRM) replaces three models: Functional Requirements for Bibliographic Records (FRBR, 1998),⁵ Functional Requirements for Authority Data (FRAD, 2008),⁶ and Functional Requirements for Subject Authority Data (FRSAD, 2010).⁷

Table 1. User tasks.

FRBR	FRAD	FRSAD	LRM
Find Identify Select Obtain	Find Identify Contextualize Justify	Find Identify Select Explore	Find Identify Select Obtain Explore

The three previous models had different, but overlapping, sets of user tasks, which were consolidated and adjusted in the “harmonized” model (LRM). See [Table 1](#) for a side-by-side comparison.

As may be seen in this side-by-side comparison, while **Find** and **Identify** are common to all the models, LRM most closely harmonizes FRBR and FRSAD. Two of FRAD’s user tasks (Contextualize and Justify) are treated as outliers, because they were formulated essentially as tasks performed by library personnel, not users in general, and thus are not included in LRM. However, as noted below, Contextualize was subsumed by the FRSAD/LRM user task **Explore**.

The “User Tasks Summary” in LRM section 3.2 describes the user tasks as follows:

... the five generic user tasks ... serve as a statement of the model’s functional scope and confirm its outward orientation to the end-user’s needs. The user tasks are phrased from the point of view of supporting the user’s ability to carry them out ...

Breaking the information seeking process down into the five generic tasks is intended to draw out each of the basic aspects of this process ... In reality information seeking is iterative and may move in a tangent at any stage. Some user tasks may happen essentially simultaneously in the user’s mind (*identify* and *select*, for example). In particular, *explore* is a separate dimension from the other tasks: in some cases providing starting points for further information seeking processes, and in others allowing browsing without any particular information goal.⁸ [Italics in original.]

For the following discussion about the LRM user tasks, each task’s definition and comment are quoted from Table 3.2 (including italics and boldface)⁹ and then briefly interpreted. Note that while each definition is about the user’s goal for each task, the “comment” includes what is needed by the information system in order for the user to accomplish the stated task or goal. These will be referred to in the discussion of the ACRL *Framework*. For the remainder of this article, when a user task is named, it will be capitalized and in boldface.

Task: Find

Definition: To bring together information about one or more resources of interest by searching on any relevant criteria.

Comment: The *find* task is about searching. The user’s goal is to bring together one or more instances of entities as the result of a search. The user may search

using an attribute or relationship of an entity, or any combination of attributes and/or relationships.

To facilitate this task, the information system seeks to **enable effective searching** by offering appropriate search elements or functionality.

In the comment for **Find**, attributes and relationships are in turn the “search elements” in the information system, more commonly referred to as the metadata (description and access points) created by catalogers. This is the meat-and-potatoes of the catalog and relies on accuracy, thoroughness, and specificity that are typically derived by catalogers from the “piece in hand” (although one may argue that this is becoming less and less the case, as with vendor records).

Task: Identify

Definition: To clearly understand the nature of the resources found and to distinguish between similar resources.

Comment: The user’s goal in the *identify* task is to confirm that the instance of the entity described corresponds to the instance sought, or to distinguish between two or more instances with similar characteristics. In “unknown item” searches, the user also seeks to recognize the basic characteristics of the resources presented.

To facilitate this task, the information system seeks to **clearly describe the resources** it covers. The description should be recognizable to the user and easily interpreted.

Accurate bibliographic description is necessary for **Identification**. “Similar resources” can be, for example, different editions of the same work or even different works with similar or identical titles. Particularly in the case of unknown item searches, this task necessarily requires subject analysis, so that the user may place a given resource into the context of the information need or research inquiry. The last sentence of the facilitation comment speaks to the need for standards for the display of the metadata.¹⁰

Task: Select

Definition: To determine the suitability of the resources found, and to be enabled to either accept or reject specific resources.

Comment: The *select* task is about reacting to possible options. The user’s goal is to make choices, from among the resources presented, about which of them to pursue further. The user’s secondary requirements or limitations may involve aspects of content, intended audience, etc.

To facilitate this task, the information system needs to **allow/support relevance judgements** by providing sufficient appropriate information about the resources found to allow the user to make this determination and act on it.

As noted in the user tasks summary quoted above from LRM, **Identify** and **Select** may happen almost simultaneously. However, distinguishing between resources is a prerequisite to determining suitability and accepting or rejecting a particular resource. Description and access points need to be sufficiently detailed, specific, and accurate, so that users may make appropriate judgments based on their information needs.

Task: Obtain

Definition: To access the content of the resource.

Comment: The user's goal in the *obtain* task is to move from consulting a surrogate to actually interacting with the library resources selected.

To fulfill this task, the information system needs to either provide direct links to online information, or location information for physical resources, as well as any instructions and access information required to complete the transaction or any restrictions on access.

Obtain is somewhat different in nature from the other four user tasks. This task is outwardly transactional and interactive, as opposed to the user-driven decision making of the other tasks, and must rely on robust technical, systems-based specifications, such as presenting links to online resources or location information and instructions on accessing materials. The action itself relies as much on the information system as on the user. Cataloging typically includes placing and maintaining working links and accurately coding the location where it is held (physical or virtual), all of which allow the user to be self-sufficient in gaining access to the **Selected** resource.

Task: Explore

Definition: To discover resources using the relationships between them and thus place the resources in a context.

Comment: The *explore* task is the most open-ended of the user tasks. The user may be browsing, relating one resource to another, making unexpected connections, or getting familiar with the resources available for future use. The *explore* task acknowledges the importance of serendipity in information seeking.

To facilitate this task the information system seeks to **support discovery** by making relationships explicit, by providing contextual information and navigation functionality.

Explore is **Find** taken to a higher level. It was originally formulated in FR SAD, but for LRM was redefined slightly to include aspects of FRAD's **Contextualize**,¹¹ so that **Explore** emphasizes the relational and contextual nature of information seeking and discovery. It goes beyond bibliographic description and relies on access points and classification that will describe

or elucidate relationships between authors, works, and subjects. These are results of what is usually referred to as subject analysis and “authority work” and are needed for collocating **Found** and **Identified** resources, such as under names (both personal and corporate), subjects, formats, genres, etc.

Framework for Information Literacy for Higher Education

The *Framework for Information Literacy for Higher Education* supersedes a previous standard, *Information Literacy Competency Standards for Higher Education* (2000). The previous standards comprised an outcomes-based model that leaned toward the “how” of information literacy. The *Framework* emphasizes more of the “what” and “why” of information literacy, which can be more solidly embedded in classroom instruction without getting mired in the mechanics of bibliographic instruction.

The *Framework* comprises six separate but interrelated “frames.” As noted in the *Framework’s* table of contents: “These six frames are presented alphabetically and do not suggest a particular sequence in which they must be learned.”¹² They are as follows:

- Authority Is Constructed and Contextual
- Information Creation as a Process
- Information Has Value
- Research as Inquiry
- Scholarship as Conversation
- Searching as Strategic Exploration

Each frame contains a brief description and conceptual commentary followed by “knowledge practices” and “dispositions.” The original texts appear in their entirety in Appendixes A through F. For the following discussion of the *Framework*, each frame’s brief description will be presented (including original boldface), followed by a general discussion of cataloging issues in the context of the given frame.

The knowledge practices and dispositions are not outcomes. Knowledge practices are intended to show ways in which learners can grow in their understanding of a given information literacy concept, while dispositions are meant to describe attitudinal or values-related perspectives encompassed by the given concept. The knowledge practices and dispositions comprise bulleted lists that are intended simultaneously to focus each frame and to allow flexibility in the classroom. They are meant to be applicable across all disciplines and information searching environments.

Authority is constructed and contextual

Information resources reflect their creators' expertise and credibility, and are evaluated based on the information need and the context in which the information will be used. Authority is constructed in that various communities may recognize different types of authority. It is contextual in that the information need may help to determine the level of authority required.

This frame is about identifying expertise and thereby judging the credibility of a source, which may be an author or a publication. That it is "contextual" means that expertise is discipline specific and that suitability depends on the information need. In the cataloging "context" we need to make some distinctions about terminology: "authority," as described in this frame, in contrast with "authority records," which are the means of recording an authorized form of name, title, or assigned term for a thing, place, time period, or concept. The two terms are related by common threads of expertise and disciplinary context.

Catalogers are usually non-experts and must rely on content in hand in order to establish and assign names and subject terminology. "Authority" as used in cataloging most commonly is derived by literary warrant, that is, the usage found in a given resource or resources (piece in hand), which may have an inherent bias. The *Framework's* own expanded commentary notes:

Experts understand the need to determine the validity of the information created by different authorities and to acknowledge biases that privilege some sources of authority over others, especially in terms of others' worldviews, gender, sexual orientation, and cultural orientations.¹³

Bias occurs, subtly or blatantly, at many levels, perhaps most familiarly at the level of individual works. However, bias also exists at broader levels, for example, as a result of collecting habits of a given institution or in the publishing habits of a given discipline, which means that viewpoints and whole groups of individuals may be excluded. Hope A. Olson noted in *The Power to Name* (2002): "Bias in the representation of marginalized groups and topics bounded by gender, race, ethnicity, nationality, class, sexuality and other factors in *LCSH* has been voiced as a concern for at least twenty-five years and has not yet been fully remedied."¹⁴ Indeed, this is still true nearly twenty years later. Bias continues to affect the perception of expertise and authority, as well as the language and terminology used to express them. The choices of names and terms will affect the ability of users to find materials to satisfy their information needs. If voices and topics are not accurately and appropriately applied to works held and thus represented in the catalog, users will not find them.

An “established authority,” in both the popular and the cataloging senses, will likely be published (often several times) and the name(s) involved will be established and recorded as access points. While this may mean that literary warrant is the most pragmatic way to establish access points in cataloging, it does not mean that literary warrant is the *only* way to determine access points. Other kinds of warrant exist. These are further discussed below in the third frame, “Information Has Value.” Authority and value are interrelated.

Information creation as a process

Information in any format is produced to convey a message and is shared via a selected delivery method. The iterative processes of researching, creating, revising, and disseminating information vary, and the resulting product reflects these differences.

In addition to information creation, this frame speaks to the ways in which information may be perceived, depending on the format in which it is presented. This is a different kind of context from that which was described above (Authority Is Constructed and Contextual), yet it also influences how a user may ascribe authority to the creator or to the resource’s content. As noted in the *Framework’s* commentary on this frame: “Experts recognize that information creations are valued differently in different contexts, such as academia or the workplace.” Two of this frame’s knowledge practices follow on this concept: learners “assess the fit between an information product’s creation process and a particular information need; [and] recognize that information may be perceived differently based on the format in which it is packaged.”¹⁵

Some information production processes, such as the manufacture or distribution of items, may be recorded in the bibliographic record, but other parts of the information/research cycle, such as peer review or methodology, are almost never acknowledged in a typical library catalog, although they may be part of description in other catalog- or index-like information environments.

Information has value

Information possesses several dimensions of value, including as a commodity, as a means of education, as a means to influence, and as a means of negotiating and understanding the world. Legal and socioeconomic interests influence information production and dissemination.

From the perspective of cataloging as a process, value is assumed: a resource presented for cataloging is meant and expected to be used.

(Consider two of Ranganathan's laws of library science: Every person his or her book. Every book its reader.) This frame asserts that contextual issues ("dimensions of value") are inherent in information and help users ascribe value to it.

The *Framework's* commentary notes that the influence of information may be used to marginalize individuals or groups: "Experts understand that value may be wielded by powerful interests in ways that marginalize certain voices"; it is re-stated in the knowledge practice by which a learner will "Understand how and why some individuals or groups of individuals may be underrepresented or systematically marginalized within the systems that produce and disseminate information."¹⁶ This is very similar to the discussion in the first frame (Authority Is Constructed and Contextual), in which authority might be used to marginalize individuals or groups and their voices and interpretations. In cataloging, this has been exemplified in the controversy over the Library of Congress subject heading "Illegal aliens."¹⁷

As noted in the discussion of authority, cataloging typically relies on literary warrant, that is, what is available in the literature at the time of cataloging to relate works to a name or a term. This does not excuse bias in cataloging but could help explain its existence. The literature itself inherently has bias: all scholarship, at some level, is interpretation, which inevitably skews in the direction of the opinion or stance (no matter the level of expertise) of the writer(s), in the language that is current *at the time of writing*. In addition to the bias that may exist in the literature itself, literary warrant can also mean that gaps in terminology may exist because literature supporting alternative views has not been published or been collected and cataloged.

As language and disciplines evolve, it is the responsibility of catalogers, subject specialists, and related agencies to keep current with acceptable terminology and usages *at the time of cataloging*, which may occur anytime from the galley proof stage (as with cataloging-in-publication programs) to many years after writing. Norman Elliott Anderson, in a chapter for *Cataloging Heresy* (1991), contended that

descriptive cataloging is an interpretive and compositional process; that the records we create should be regarded as living, not static; that many of the elements of descriptive cataloging are non-neutral in several different ways, some good and some not good; and that we ought to take a new look at how we handle the mix of universality and non-neutrality in our descriptive cataloging.¹⁸

To riff on Ranganathan again: Not only is the library a growing organism — so is the catalog.

This frame and its knowledge practices are very closely related to the concepts of "cultural warrant" and "user warrant," the former of which has been described by Beghtol (2005) and the latter in the *Guidelines for*

the Construction, Format, and Management of Monolingual Controlled Vocabularies (ANSI/NISO Z39.19-2005 (R2010)). Beghtol defined cultural warrant as follows:

The personal and professional cultures of information seekers and information workers warrant the establishment of appropriate fields, terms, categories, or classes in a knowledge representation and organization system. Thus, cultural warrant provides the rationale and authority for decisions about what concepts and what relationships among them are appropriate for a particular system. This warrant traditionally arises from the presumed information needs of the potential users of the system.¹⁹

Thus, the catalog itself is a product of culture and research. Beghtol elaborated a definition of culture:

In general, culture encompasses the various shared values, beliefs, histories, and activities of a cohesive group of people. This broad description of culture means that ... a user's culture can be relatively broad (e.g., a national, religious, or ethnic culture) or relatively narrow (e.g., an academic, artistic, or institutional discipline).²⁰

The National Information Standards Organization (NISO) defined user warrant as “the language of users” and further elaborated on it as being “generally reflected by the use of terms in requests for information on the concept or from searches on the term by users of an information storage and retrieval system.”²¹

Every institution, and thus any catalog it produces, exists within a particular community, which has its own culture, its own audience. Its users will be a mix of novices and experts who are learning and/or reinforcing language norms (“the language of the users”) that will in turn influence the success (or failure) of the catalog in meeting their information needs. Catalogers need to engage in research (beyond that required for literary warrant processes) relative to the disciplinary or cultural expectations of the catalog's users, in order for bibliographic descriptions to be optimally effective for directing users to relevant resources.

Research as inquiry

Research is iterative and depends upon asking increasingly complex or new questions whose answers in turn develop additional questions or lines of inquiry in any field.

This frame deals with the research process itself, as opposed to the results or products created by the process that were discussed in the second frame (Information Creation as a Process). Here the focus is on the flow from first formulating a research question (that is, identifying an information need) to subsequently organizing, investigating, analyzing, summarizing, and interpreting information as it is found and then synthesizing it into

new knowledge. This includes the pre-thinking and designing that occurs before the user even approaches the catalog. However, Creswell and Creswell consider the library to be the first step in any research project; they discuss how to use the library and databases to conduct literature reviews in order to answer the fundamental question: *should* the research be done.²² Holdings will help determine if a question has been treated by others, in what way, and to what extent. The research/inquiry process is iterative, and the catalog-related user tasks are an integral part of the process.

To facilitate the user's search and exploration for resources to inform the research process, it is necessary (as noted in the discussions for the first and third frames, *Authority Is Constructed and Contextual*, and *Information Has Value*, respectively) for the description and subject analysis of resources to be as complete and as accurate as possible. This particularly includes the need for consistent application of current terminology. As discussed in the third frame, terms change over time, and it becomes incumbent on the cataloging agency to stay abreast of those changes so that relevant resources can remain together (physically and virtually) as well as discoverable.

Scholarship as conversation

Communities of scholars, researchers, or professionals engage in sustained discourse with new insights and discoveries occurring over time as a result of varied perspectives and interpretations.

This frame extends the first three frames (*Authority Is Constructed and Contextual*; *Information Creation as a Process*; *Information Has Value*), as summarized in the frame's accompanying commentary: "While novice learners and experts at all levels can take part in the conversation, established power and authority structures may influence their ability to participate and can privilege certain voices and information."²³ This is recast in the frame's last disposition, in which learners "recognize that systems privilege authorities and that not having a fluency in the language and process of a discipline disempowers their ability to participate and engage."²⁴ In the catalog, those authority structures may be manifested in detailed and well-researched access points (supported by authority records with cross-references and other distinctive authority-related information). A liberal use of such access points builds relationships between works, creators, concepts, and terms, thereby supporting or influencing users' perceptions of those structures and consequently informing users' choices of resources.

This circles back to the second frame (*Information Creation as a Process*), which emphasizes the point that knowledge creation (and, one

might add, its subsequent discovery) may occur in any format. Scholarship is an ongoing discourse:

Developing familiarity with the sources of evidence, methods, and modes of discourse in the field assists novice learners to enter the conversation. New forms of scholarly and research conversations provide more avenues in which a wide variety of individuals may have a voice in the conversation.²⁵

This is a point made by van Arnhem and Finch in a discussion of visual literacy (the use of images and non-textual resources) in this frame (Scholarship as Conversation) in which learners

understand that scholarly conversations and information related to their topics come from multiple information landscapes, each with varying copyright, intellectual property, and fair use rules as well as standards for citing and attribution.²⁶

As noted by van Arnhem and Finch, “attribution to relevant previous research is also an obligation of participation in the conversation. It enables the conversation to move forward and strengthens one’s voice in the conversation.”²⁷ This extends to the catalog itself — full and accurate attributions are access points, the bibliographic metadata that is crucial for the discovery of resources in the catalog. The same basic elements in a writer’s bibliographical citations are all present in a well-described bibliographic record: author, title, place of publication, publisher, and year. When any of these elements is missing from a citation or from a bibliographic record, valuable distinctive information is lost and unavailable for furthering the establishment of a voice in the conversation, either that of the author cited or the author doing the citing.

In the cataloging world, a “lack of fluency” is a particular danger for a non-expert cataloger trying to be as specific as possible with subject analysis and classification. A misunderstanding of the subject of a work can lead to the application of incorrect terms and subsequently to misclassification, making the resource difficult, if not impossible, to find because it has not been correctly placed in context with other related resources.

Searching as strategic exploration

Searching for information is often nonlinear and iterative, requiring the evaluation of a range of information sources and the mental flexibility to pursue alternate avenues as new understanding develops.

It is fortuitous that this frame is presented last in the *Framework*. In the bibliographic environment it is the culmination of the thoughts presented in the previous five frames. This is where the user meets the catalog.

However, in the context of information literacy, searching is a broad process and encompasses all of the LRM user tasks. The *Framework’s*

commentary opens with a statement that connects to **Find**, **Identify**, **Select**, and **Obtain**; **Explore** is in the name of the frame itself.

The act of searching often begins with a question that directs the act of finding needed information. Encompassing inquiry, discovery, and serendipity, searching identifies both possible relevant sources as well as the means to access those sources.²⁸

Searching is an iterative process for meeting an information need. In this frame (as opposed to the fourth frame, Research as Inquiry), the search itself (not the research question) is the focus. Searching requires strategy and a willingness to change direction as new information and understanding are uncovered. Context within a discipline (as well as the scope of a given inquiry) is usually needed so that the user may adequately evaluate the relevance of sources. As noted in the discussion of Research as Inquiry, accurate and comprehensive description, and subject analysis reflected in sufficient and appropriate access points, are needed for successful searching.

The second knowledge practice (“identify interested parties... who might produce information... and then determine how to access that information”²⁹) is a restatement that rolls together **Find**, **Identify**, **Select**, and **Obtain**. Bibliographic records must contain the necessary metadata for elucidating relationships between works, creators, concepts, and so on, which relies on exhaustive authority work and consistent application of names and terms.³⁰ Further for the successful accomplishment of this knowledge practice, publication and access-related information must be fully and accurately represented in the catalog record.

The LRM user tasks informed by the *Framework*

Given the context in which the LRM user tasks are defined (that is, in a high-level cataloging model), it would appear that they would mainly pertain to the “Searching as Strategic Exploration” frame. It is posited in this discussion that cataloging and the user tasks have relevance to each frame and may inform (and be informed by) the *Framework*’s concepts, knowledge practices, and dispositions, thereby having the potential to create an environment capable of enriching users’ experiences with the catalog (or any catalog-like information environment) and meeting those users’ information needs. Cataloging can support information literacy and, conversely, the tenets of information literacy can support decision-making in cataloging.

Find

Authority is constructed and contextual

Find involves searching on names and terms, as well as on known titles and/or keywords. Do names and terms reflect expertise (authority) and

current disciplinary vocabularies? Are there sufficient cross-references and other relational tools in the catalog to guide the user to relevant resources? Are names and terms consistently applied across resources, regardless of format, so that they may be pulled together in search results? While this frame is primarily about the authority of persons, those persons' expertise is typically discipline or subject specific. Thus, authority can extend to subject access, by means of appropriate and current terminology, which is crucial for helping novices and expert researchers alike to navigate the catalog and obtain access to resources to meet information and learning needs.

Further, variant names for individuals need to be identified, recorded, and cross-referenced to bring works together; some individuals, however, change their identities, which may be tied to different bodies of works—how does the cataloger acknowledge the connections and still honor the separations? Subject terms change and evolve over time, recognition of which necessitates understanding whether a search term relates to an already-established concept, and thus requires an update to an existing authority record, or whether a term is indeed a new concept that needs to be freshly established and recorded.

Information creation as a process

With its emphasis on the fit of format and content of an information resource to a searcher's information need, this frame maps more completely to **Identify**. However, occasionally format may be a primary aspect of a search, such as a sound or video recording of a particular musical work (as opposed to a score). Successful **Finding** requires that elements such as format and edition are present in the description.

Information has value

The **Findability** of a work is directly impacted by the currency, appropriateness, and accuracy of a work's description and access points, because the bibliographic record stands in for the work itself in the catalog. If a discipline's or user community's current terminology is not included in the catalog, a user's ability to **Find** relevant materials is considerably hindered. As noted above in the discussion of this frame, literary warrant may not be sufficient for providing the necessary and appropriate description or subject analysis of resources for optimal searching. Research beyond the piece in hand may be needed, such as searching the cataloger's own catalog for previous treatment of similar works (which may prompt remediation or other updating) or seeking subject or discipline expertise from

colleagues or the catalog's user base (which are expressions of both cultural and user warrant).

Research as inquiry

Research, as noted in this frame, is an iterative process; inquiry is the focusing mechanism for that process. Although this frame is not about searching *per se*, the **Find** user task is a necessary step in the pursuit of information (as is **Explore**), especially to provide context for the inquiry itself. Thorough and accurate provision of access points and consistent subject analysis will help users **Find** the information they need. If anything, Research as Inquiry can also serve as a guiding principle for the authority work (which itself is research) that is necessary for establishing names and cross-references and for maintaining the currency of controlled vocabularies.

Scholarship as conversation

Users must be able to **Find** resources by using terms that are relevant to the topic for which they are seeking information. Are there sufficient and accurate access points to reflect the nature of a resource and consequently allow it to collocate with like materials in all formats? Do controlled terms accurately reflect current terminology? Do older materials collocate with more recent resources? Are non-textual and non-print resources treated consistently and with the same care as print/textual works? In this frame, learners must be able to “summarize the changes in scholarly perspective over time on a particular topic within a specific discipline,”³¹ which finds its cognate in the catalog as the need for continual maintenance of controlled vocabularies to reflect those changes. In a sense, catalogers are in ongoing conversation with the catalog by engaging in the research needed for up-to-date authority work.

Searching as strategic exploration

This is the only frame in which the knowledge practices explicitly mention a catalog-related concept: the learner should be able to “use different types of searching language (e.g., controlled vocabulary, keywords, natural language) appropriately.”³² These, of course, are foundational to **Find** and **Explore**; the catalog needs to support a variety of search languages (as so succinctly noted in the knowledge practices). This can be done only if the bibliographic description, access points, and subject analysis are accurate, current, specific, thorough, and appropriate. Creation and maintenance of authority records for names and terms are necessary for this level of quality to be accomplished.

Identify

Authority is constructed and contextual

Identify is a decision-making action of distinguishing between **Found** resources based on their characteristics. If subject or keyword terms are the basis of an initial search, authorized forms of names may be used to sort and to make decisions on which materials to use. If names are used for a search, subjects and classification may be used to collocate sources for decision making. As noted above, have names and terms been consistently applied?

Information creation as a process

Cataloging as typically practiced in libraries makes a distinction in the package: monographs and serials (as a whole) are described; chapters and individual articles are not. Description and subject analysis of non-textual works and non-print media, such as maps, scores, audio or video recordings, and born-digital resources, tend to be problematic as well. Functions associated with this frame (such as assessing “fit” or “matching an information need with appropriate product”³³) map to the **Identify** user task. Elements such as format, edition, place of publication, and publisher, are needed for successful **Identification** and sorting of **Found** resources.

Of further note is the knowledge practice, “recognize the implications of information formats that contain static or dynamic information.” This points to the necessity of accurately distinguishing in the catalog between not only monographic and continuing resources, but also between serials and integrating resources.

Harkening back to the first frame (Authority Is Constructed and Contextual), some publishers and information-providers are considered more “authoritative” or desirable than others in various disciplines. Consequently, for comprehensive **Finding** and **Identification**, publication and distribution information need to be recorded, in addition to format. Ideally, all publishers and distributors would be established and supported by authority records that could provide context for the type of information products supplied and to what audience. Many have been, but this is not a consistent practice, nor is the established information evenly or consistently applied.

Information has value

Novice researchers may not yet be familiar with the history of a discipline and thus may not be able to call on relevant terms. Current terms, and accurate cross-references leading to them, are necessary so that a user can get to a record for a relevant resource. Literary warrant, the typical basis

for establishing standardized terms and forms of name, may not be sufficient for keeping up with current language in a given discipline. Other types of warrant (such as cultural warrant and user warrant) provide pathways by which vocabularies can be maintained outside the realm of collected publications and into the world of the users themselves. Without a clear relationship to the current state of scholarship in a discipline (or to the collecting habits of a supporting institution), a resource's description may not be (or may no longer be) accurate or thorough enough to be **Identifiable** as meeting the user's information need.

Research as inquiry

Identification is the process by which the user further contextualizes and makes distinctions among **Found** resources, noted in a knowledge practice as being able to "organize information in meaningful ways." As noted in the *Framework's* commentary on this frame, "Novice learners acquire strategic perspectives on inquiry and a greater repertoire of investigative methods."³⁴ This is fully successful in the library context only with the provision of high-quality metadata.

Scholarship as conversation

As noted in the comment on this frame, scholarship is an ongoing discourse that evolves over time. In cataloging, as noted above in **Find**, this is reflected in controlled vocabularies that are kept current by the addition of new terms and the editing of older ones with cross-references as they enter or change within the language of a discipline. Well-formed access points, in combination with sufficient and accurate description and publication information, maximize the opportunities that users will have to successfully **Identify** resources that they need.

Searching as strategic exploration

The user's **Identification** and sorting of **Found** resources is guided by the research inquiry (information need). This is addressed in several knowledge practices and dispositions, in which learners

- identify interested parties, such as scholars, organizations, governments, and industries, who might produce information about a topic...
- design and refine needs and search strategies as necessary, based on search results
- manage searching processes and results effectively

- understand that first attempts at searching do not always produce adequate results
- recognize the value of browsing and other serendipitous methods of information gathering
- persist in the face of search challenges, and know when they have enough information to complete the information task³⁵

All of these practices and dispositions rely on quality metadata, consistently applied.

Select

Authority is constructed and contextual

The goals of the Authority frame are needed for **Find** and **Identify**, as discussed above in their respective sections. **Select** is the user's action, the decision-making, that follows **Identification** of **Found** resources.

Information creation as a process

Creation processes tend to be discipline specific: humanities generally favor monographs, while the sciences tend to favor articles,³⁶ yet all formats are used by all disciplines, just in different patterns. Intner noted that "medium determines, or, at least, influences, potential perceptions of informational messages, and organizational structures and systems might be expected to recognize and address the resulting differences."³⁷ She further wrote that "knowledge of material use patterns is an important feedback mechanism for catalogers to evaluate the efficacy of their cataloging, indexing, and classification systems."³⁸ Ideally, catalogers would communicate closely with their colleagues in public services in order to understand users' information needs and, if possible, to ameliorate problems caused by the structures of both the catalog (generally) and standards in the creation of its records (specifically).

As novices learn the lingo of their disciplines on the way to becoming experts, they begin to recognize names and to prefer particular formats for meeting their information needs; they consequently learn to **Select** discipline-appropriate materials. However, without such cues in the catalog, resources may be overlooked or unrecognized.

Information has value

Granularity beyond current access points for names and subjects is often necessary for a user to **Select** any **Identified** resources for further

investigation. Such level of description may include summaries or tables of contents, additional publication information such as reprint or series information, or notes about genre or the intended audience. A lack of these types of information may contribute to underrepresentation of individuals or groups within a discipline or research area, and limit or prevent **Selecting** and **Obtaining** a perfectly good resource.

Research as inquiry

The decision making involved in **Selection** is based on the research questions that the user is pursuing and the results that were sorted and organized in the process of **Identification**. The level of granularity in the bibliographic description can support the knowledge practice in this frame in which learners develop an ability to “monitor gathered information and assess for gaps or weaknesses.”³⁹ Such assessment will likely lead to **Selection** of some resources over others.

Scholarship as conversation

Besides assessing gaps in research, learners also establish knowledge practices, as noted in this frame, to “critically evaluate contributions made by others in participatory information environments; [and to] recognize that a given scholarly work may not represent the only or even the majority perspective on the issue.”⁴⁰ These practices are necessary for appropriately **Selecting** materials from a list of **Identified** resources. Especially in the latter case, in which a work may represent a variant perspective, well-formed accurate metadata will aid in such **Selection**.

Searching as strategic exploration

The user’s **Selection** of resources ultimately will be dictated by the information need or research inquiry. This is reinforced in the third disposition (which harkens back to the second frame, Information Creation as a Process), in which learners “realize that information sources vary greatly in content and format and have varying relevance and value, depending on the needs and nature of the search.” However, they may be hindered if a bibliographic record does not contain sufficient information to make nuanced decisions. Enhancements to records, such as additional description and access points (especially for non-textual or archival materials), incur costs in time and money (that is, paying someone to make those enhancements) that may not outweigh the benefits, depending on the type and size of the library and its available financial resources.

Obtain

Authority is constructed and contextual

Obtain in the LRM sense is not addressed in this frame. It is the transaction that results from positive **Identification** and **Selection** of a resource based on its description and access points.

Information creation as a process

Because of the differential granularity of the description of formats in the catalog, the **Obtain** task may be more difficult or time-consuming for users, especially when trying to access specific chapters or articles or non-textual/non-print works (which may require a specific type of device for viewing). This issue may be somewhat mitigated by the addition of tables of contents in bibliographic records, but this is not a universal or consistent practice. A different route (with varying levels of success) has been the building out of discovery layers that search across the catalog and some full-text databases and indexes. However, even in these cases, different vocabularies and granularities of description may lead to less-than-optimal results.

Information has value

This frame relates directly to the **Obtain** task by acknowledging that an information literate researcher will “recognize issues of access or lack of access to information sources.”⁴¹ In the catalog environment, currency of a different sort is required for this task: To **Obtain** a resource means that various types of location information (and any restrictions to physical or digital access) must be up to date, from place of publication or distribution to the location of a physical item within the institution’s collection to a URL for an online work.

Research as inquiry

Obtain in the LRM sense is addressed obliquely in the disposition in which learners “follow ethical and legal guidelines in gathering and using information.”⁴²

Scholarship as conversation

Obtain in the LRM sense is not addressed in this frame.

Searching as strategic exploration

The concept that maps to **Obtain** is referred to as “access” in the *Framework*. It is expressed in the knowledge practice whereby learners “understand how information systems (i.e., collections of recorded information) are organized in order to access relevant information.”⁴³ Such organization depends on the consistent application of metadata, including information related to publication and location information for physical and digital resources, as well as current instructions and related information for online materials.

Explore

Authority is constructed and contextual

Two dispositions in this frame most closely represent **Explore**, those in which learners “develop and maintain an open mind when encountering varied and sometimes conflicting perspectives; [and] motivate themselves to find authoritative sources, recognizing that authority may be conferred or manifested in unexpected ways.”⁴⁴ As authority (expertise) is necessarily conferred and inferred in relationship to a discipline or subject area, accurate and thorough subject analysis and the establishment of authorized names and terms, with sufficient and appropriate cross-references, must be done to enable full and consistent discovery of relevant resources. In the cataloging world this can be accomplished only by “authority work”; successful **Exploration** requires the same relational concepts as the **Find** user task.

Information creation as a process

The disposition that states that learners “value the process of matching an information need with an appropriate product”⁴⁵ is relevant to **Explore** (in addition to its value for **Identify**, discussed above). To serve users well, the records **Found** must contain accurate descriptions in order to bring out relationships among resources and consequently to achieve a needed match. These tenets of relationship undergird **Explore** and support the last knowledge practice of this frame, so that learners “develop... an understanding that their choices impact the purposes for which the information product will be used and the message it conveys,”⁴⁶ which can be interpreted as “know your audience and write [or record or perform] for them” — perhaps the ultimate driver and context for the creation, dissemination, and use of new knowledge.

Information has value

Explore, with its emphasis on relationships between works and their creators, is constrained as described above in the discussion around **Finding** relevant materials. However, as noted in the *Framework's* discussion of this frame: "value may...be leveraged by individuals and organizations to effect change and for civic, economic, social, or personal gains."⁴⁷ This milieu comprises a multi-faceted context for **Exploration** through which users must navigate to **Find** resources that they feel comfortable using. Thorough, accurate, and consistent treatment of materials will help elucidate the relationships between works and their producers so that users may recognize the value of what they are **Finding** and subsequently expand their opportunities for meeting information needs.

Research as inquiry

The essence of this frame is **Exploration** to meet an information need. The first disposition makes this explicit: Learners "consider research as open-ended exploration and engagement with information."⁴⁸ The emphasis in this frame is not on the search, which is dealt with in the sixth frame (Searching as Strategic Exploration). Rather, it is on the research question itself (that is, the information need) and the process that leads to new knowledge. This harmonizes with the commentary from the LRM "User Tasks Summary" that was quoted at the beginning of this article, which called out **Explore** as "providing starting points for further information seeking processes."⁴⁹ The relationships built by the metadata that support **Find** are the drivers behind further **Explorations** that expand research and create knowledge.

Scholarship as conversation

The ongoing nature of "scholarship as conversation" necessitates currency and nuance in controlled vocabularies, which is necessary for building the bibliographic relationships that support continued **Exploration**. Leach noted that in a way, "the library catalog and its records are in conversation with library users."⁵⁰ She further elaborated that as this conversation occurs, "members of the cataloging community are rarely present with local community members to explain their decisions."⁵¹ Yet those decisions yield metadata that may help or hinder users in their search for resources. Catalogers need to engage in conversations with public services colleagues and users, so that catalogers may receive useful feedback from the communities they (and the catalog) serve. This echoes Intner's position (discussed in the second frame, Information Creation as a Process), in which

catalogers need feedback on material use so they can evaluate and adjust cataloging practices (such as authority work and classification) as needed in order to best serve their community. Such feedback would enable practical decisions that “are made on the basis of user needs for particular types of information and are consistent with the expected behaviors of members of the community in searching, retrieving, and using the information.”⁵²

Searching as strategic exploration

The fifth knowledge practice, “design and refine needs and search strategies as necessary, based on search results,” while relevant to **Identify**, is also pertinent to **Explore**, as both user tasks rely on not just accuracy but consistency in the application of access points, especially subject headings, for relating and pulling together resources. This is especially challenging with multi- or interdisciplinary resources but speaks to the need for liberal use of subject headings and other relevant access points. Indeed, the **Explore** user task is at the root of this knowledge practice and the whole frame. The iterative and often circuitous path of research must be supported by metadata that will allow the user to uncover the full “aboutness” of and relationships between works. This can only be done by accurate description and specific access points (established forms of names and current subject terminology, all with comprehensive cross-references as needed).

Conclusion

This discussion has been on ways in which cataloging, as exemplified by the LRM user tasks, can support, and be informed by, information literacy practices for meeting users’ information needs. The six frames of the *Framework for Information Literacy for Higher Education* have offered different perspectives and context by which to examine various aspects of the five user tasks and their implications for cataloging practices, that is, bibliographic description and subject analysis.

Table 2 summarizes the intersections of the user tasks and the frames in the form of guiding questions for cataloging outcomes. What has emerged from this examination is that information literacy and cataloging rely on relationships: between works, between works and their creators, and perhaps most importantly, between works and their users. Cataloging can and must support information literacy. In the catalog these relationships can be brought out by careful description and respectful, well-researched access points.



Table 2. Guiding questions: intersections of user tasks with frames.

Find		Identify		Select		Obtain		Explore	
Authority Is Constructed and Contextual	Do names/terms reflect expertise? Are variants of names cross-referenced accurately? Are names/terms current and respectful?	Do works by the same author, or works on similar subjects, collocate together?	(See Find and Identify intersections with Authority Is Constructed and Contextual.)	(Not applicable.)	Are names/terms fully established? Have they been thoughtfully and consistently applied so users can Explore relationships between works?				
Information Creation as a Process	Searchers may be looking for a specific edition or format. Are these elements accurately described?	Can the user distinguish between Found resources?	Are descriptions specific enough and relevant, so that the user can Select materials for use with confidence that they are getting what they need?	Is a device needed for accessing the Selected work? Are there other aspects of the work that may hinder or affect its use? Is this information accurately described?	(See Identify and Select intersections with Information Creation as a Process.)				
Information Has Value	Is a user community's current terminology reflected in bibliographic records or the catalog?	Are description and subject analysis granular enough to accurately represent a work? Is publication information fully provided? Do terms reflect current usage in the context of a discipline or relevant community of use, so that differences between Found resources may be distinguished?	(See Find and Identify intersections with Information Has Value.)	(See above intersection of Obtain with Information Creation as a Process.)	See the intersection between Identify and Information Has Value. Are the descriptions thorough enough to allow the user to see relationships among Found works?				
Research as Inquiry	Are terms consistently and accurately applied? Has subject analysis been thorough? Are terms up to date?	Will the user be able to make sense of Found resources based on the description, within the context of the user's information need? See above intersection of Identify with Information Has Value.	(See intersections of Identify with Information Has Value and Research as Inquiry.)	Is the catalog providing legal access to the information described?	Are names/terms consistently applied so users can Explore relationships between works?				
Scholarship as Conversation	Do older materials collocate with recent resources? Have non-textual works received the same treatment (description/subject analysis) as textual related works? Are controlled vocabularies up to date?	Have access points been clearly constructed and consistently applied, so that users may distinguish between similar resources?	Are subject analysis and resulting access points specific and representative, so that after differentiating between resources, the user can confidently Select materials that they need?	(Not applicable.)	Do controlled vocabularies contain fully established terms with cross-references? Do catalogers understand their user base well enough to be able provide necessary and relevant descriptions and subject access?				
Searching as Strategic Exploration	Are bibliographic description, access points, and subject analysis accurate, current, specific, thorough, and appropriate?	Are there enough specific access points in a description so that the user may be able to refine a search?	(See Find and Identify intersections with Searching as Strategic Exploration.)	Is the organization of the information clear, accurate, and up to date, so that the user can understand how to get access (physical or digital) to the Selected resource?	Do descriptions contain liberal consistent use of established names and up-to-date terms and thorough description?				

Another way to put it is that context matters. The *Framework's* commentary on Search as Strategic Exploration, notes “that information searching is a contextualized, complex experience that affects, and is affected by, the cognitive, affective, and social dimensions of the searcher.”⁵³ All users have unique and individual information needs, which are informed by various and differing research contexts, such as discipline area, level of expertise, or purpose of a specific research question, and by personal characteristics and background, such as ethnicity, gender, nationality, language, etc. Any of these may lead to situations in which users may feel marginalized or unheard, or conversely, to situations in which users are empowered to learn and grow as their information needs are met. The descriptions and vocabularies used in cataloging create relationships between works, creators, concepts, and users; the language used to describe those relationships may hinder the user if personal and disciplinary differences are not respected and accounted for. This point was emphasized in the *Framework* particularly in Authority is Constructed and Contextual and Information Has Value. However, while it may be newly placed within this space of information literacy, as may be inferred in those frames' respective discussions, it is not new to the sphere of cataloging. Olson and Schlegl wrote that

... users must not be regarded as homogeneous. The danger of speaking of “the user” is that that phrase suggests a particular user, a “majority reader,” rather than the heterogeneous groups of users that vary from one context to another. ... standards have value ... but they lose their effectiveness if they are not carefully and equitably applied. Careful application requires that a standard be fully and accurately utilized. Equitable application requires adaptation to local context, whether “local” is at the institutional, national or cultural level.⁵⁴

In cataloging, this means a cultural and disciplinary sensitivity to terminologies and forms of name. Context in cataloging is thus built by the relationships that emerge from authority work —respectfully establishing forms of name and maintaining controlled vocabularies and classifications for providing acceptable and up-to-date access points, so that users will be able to pull together (**Find**, **Identify**, **Select**) and access (**Obtain**) relevant materials to meet their information needs, and to further **Explore** other questions as they arise in their research.

From this context-related position, we must be willing to tailor bibliographic descriptions in a manner that will meet users' expectations. Doing so will allow novice and expert searchers alike to be able to **Find**, **Identify**, **Select**, **Obtain**, and further **Explore** the catalog within and outside the boundaries of their disciplines. To engage in this kind of bibliographic practice could lead back to issues of cultural and user warrant, as opposed to the current practice of literary warrant. To go beyond literary

warrant would require conversations not only between catalogers and the user communities in which they are situated, but also between the catalogers and the catalogs they maintain and enlarge. Catalogers are users, too, and benefit as well by the user tasks in all their intersections with information literacy. Such potential conversations would ultimately enrich the catalog, to enhance (perhaps even reclaim?) its position as a primary tool for research and lifelong learning.

ORCID

C. Rockelle Strader  <http://orcid.org/0000-0001-8518-7078>

Notes

1. Association of College and Research Libraries. *Framework for Information Literacy for Higher Education* (Chicago: Association of College and Research Libraries, 2016), <http://www.ala.org/acrl/sites/ala.org/acrl/files/content/issues/infolit/framework1.pdf>, 29. [Hereafter cited as “Framework”]
2. *Framework*, 8.
3. Pat Riva, Patrick Le Boeuf and Maja Žumer, “IFLA Library Reference Model: A Conceptual Model for Bibliographic Information” (2017), 15, https://www.ifla.org/files/assets/cataloguing/frbr-lrm/ifla-lrm-august-2017_rev201712.pdf [Hereafter cited as “LRM”]
4. Erin Leach. “Using the *Framework* to Frame: Cataloging Policy and Practice as Seen through the Lens of *The Framework for Information Literacy for Higher Education*,” in *At the Helm: Leading Transformation: The Proceedings of the ACRL 2017 Conference, March 22-25, 2017, Baltimore, Maryland*, ed. Dawn M. Mueller (Chicago: Association of College and Research Libraries, 2017), 719.
5. IFLA Study Group on the Functional Requirements for Bibliographic Records, “Functional Requirements for Bibliographic Records: Final Report” (1998, 2009), <https://www.ifla.org/publications/functional-requirements-for-bibliographic-records>.
6. IFLA Working Group on Functional Requirements and Numbering of Authority Records (FRANAR), “Functional Requirements for Authority Data: A Conceptual Model” (2008, 2013), https://www.ifla.org/files/assets/cataloguing/frad/frad_2013.pdf.
7. Marcia Lei Zeng, Maja Žumer, Athena Salaba, eds., “Functional Requirements for Subject Authority Data (FRSAD): A Conceptual Model” (2010), <https://www.ifla.org/files/assets/classification-and-indexing/functional-requirements-for-subject-authority-data/frsad-final-report.pdf>.
8. LRM, 15.
9. LRM, 16.
10. See: International Federation of Library Associations and Institutions, ISBD Review Group, *ISBD: International Standard Bibliographic Description*, consolidated ed., IFLA Series on Bibliographic Control, 44 ((Berlin/Münich De Gruyter Sauer, 2011).
11. Pat Riva, Patrick Le Boeuf and Maja Žumer, “Transition Mappings: User Tasks, Entities, Attributes, and Relationships in FRBR, FRAD, and FRSAD Mapped to Their Equivalents in the IFLA Library Reference Model” (2017), 2, <https://www.ifla.org/files/assets/cataloguing/frbr-lrm/transitionmappings201708.pdf>.

12. *Framework*, 5.
13. *Framework*, 12. (See also Appendix A.)
14. Hope A. Olson, *The Power to Name: Locating the Limits of Subject Representation in Libraries* (Dordrecht, Netherlands: Kluwer Academic Publishers, 2002), 143.
15. *Framework*, 14. (See also Appendix B.)
16. *Framework*, 16. (See also Appendix C.)
17. For a history of this controversy and steps being taken by some in the library community, see Association for Library Collections and Technical Services, Subject Analysis Committee, Report of the SAC Working Group on Alternatives to LCSH "Illegal aliens", accessed January 18, 2021, <https://alair.ala.org/handle/11213/14582>.
18. Norman Elliott Anderson, "The Non-Neutrality of Descriptive Cataloging," *Cataloging Heresy: Challenging the Standard Bibliographic Product*. Proceedings of the Congress for Librarians, February 18, 1991, St. John's University, Jamaica, New York, ed. Bella Hass Weinberg (Medford, N.J.: Learned Information, 1992), 16.
19. Clare Beghtol, "Ethical Decision-Making for Knowledge Representation and Organization Systems for Global Use," *Journal of the American Society for Information Science and Technology* 56, no. 9 (2005): 904.
20. Beghtol, "Ethical Decision-Making," 904.
21. National Information Standards Organization, *Guidelines for the Construction, Format, and Management of Monolingual Controlled Vocabularies* (ANSI/NISO Z39.19-2005 (R2010)). (Baltimore: National Information Standards Organization, 2010), 16.
22. John W. Creswell and J. David Creswell, *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*, 5th ed. (Los Angeles: SAGE Publications, 2018), 23-34.
23. *Framework*, 20. (See also Appendix E.)
24. *Framework*, 21. (See also Appendix E.)
25. *Framework*, 20. (See also Appendix E.)
26. Jolanda-Pieta van Arnhem and Jannette L. Finch, "Pulling It All Together: Sharing Your Research Story" in *Scholarship as Conversation (Framing Information Literacy: Teaching Grounded in Theory, Pedagogy, and Practice*, vol. 5. Publications in Librarianship, No. 73, ed. Mary K. Oberlies and Janna Mattson), 443.
27. Van Arnhem and Finch, 503.
28. *Framework*, 22. (See also Appendix F.)
29. *Ibid.*
30. For a treatment of how bibliographic relationships may be facilitated and expanded by entity-relationship models such as FRBR and LRM, see: Hope A. Olson, "How We Construct Subjects A Feminist Analysis," *Library Trends* 56, no. 2 (Fall 2007): 509-541.
31. *Framework*, 21. (See also Appendix E.)
32. *Framework*, 22. (See also Appendix F.)
33. *Framework*, 14-15. (See also Appendix B.)
34. *Framework*, 18. (See also Appendix D.)
35. *Framework*, 22-23. (See also Appendix F.)
36. See, for example: Wolfgang Glänzel and Urs Schoepflin, "A Bibliometric Study of Reference Literature in the Sciences and Social Sciences," *Information Processing & Management* 35, no. 1 (1999): 31-44; Anton J. Nederhof, "Bibliometric Monitoring of Research Performance in the Social Sciences and the Humanities: A Review," *Scientometrics* 66, no. 1 (2006): 81-100; C. Rockelle Strader, "Citation Analysis: Do Age and Types of Materials Cited Correlate with Availability of Appropriate Library

- of Congress Subject Headings?” *Library Resources and Technical Services* 56, no. 4 (2012): 238–253.
37. Sheila S. Intner, “Rejecting Standard Cataloging Copy: Implications for the Education of Catalogers,” in *Cataloging Heresy: Challenging the Standard Bibliographic Product*, Proceedings of the Congress for Librarians, February 18, 1991, St. John’s University, Jamaica, New York, ed. Bella Hass Weinberg (Medford, N.J.: Learned Information, 1992), 124.
 38. Intner, “Rejecting Standard Cataloging Copy,” 125.
 39. *Framework*, 18. (See also Appendix D.)
 40. *Framework*, 20–21. (See also Appendix E.)
 41. *Framework*, 16. (See also Appendix C.)
 42. *Framework*, 19. (See also Appendix D.)
 43. *Framework*, 22. (See also Appendix F.)
 44. *Framework*, 13. (See also Appendix A.)
 45. *Framework*, 15. (See also Appendix B.)
 46. Ibid.
 47. *Framework*, 16. (See also Appendix C.)
 48. *Framework*, 19. (See also Appendix D.)
 49. LRM, 15.
 50. Leach, “Using the *Framework* to Frame,” 722.
 51. Leach, “Using the *Framework* to Frame,” 723.
 52. Intner, “Rejecting Standard Cataloging Copy,” 126.
 53. *Framework*, 22. (See also Appendix F.)
 54. Hope A. Olson and Rose Schlegl, “Bias in Subject Access Standards: A Content Analysis of the Critical Literature,” in *Information Science Where Has It Been, Where Is it Going?* Proceedings of the 27th Annual Conference of the Canadian Association for Information Science / *Les sciences de l’information: D’où viennent-elles et où s’en vont-elles?* Actes du 27e Congrès annuel de l’Association canadienne pour les sciences de l’information, Université de Sherbrooke, June 9–11, 1999, ed. James Turner, accessed January 18, 2021, <https://doi.org/10.29173/cais460> 246.

APPENDIX A: AUTHORITY IS CONSTRUCTED AND CONTEXTUAL (*Framework*, pp. 12–13)

Information resources reflect their creators’ expertise and credibility, and are evaluated based on the information need and the context in which the information will be used. Authority is constructed in that various communities may recognize different types of authority. It is contextual in that the information need may help to determine the level of authority required.

Experts understand that authority is a type of influence recognized or exerted within a community. Experts view authority with an attitude of informed skepticism and an openness to new perspectives, additional voices, and changes in schools of thought. Experts understand the need to determine the validity of the information created by different authorities and to acknowledge biases that privilege some sources of authority over others, especially in terms of others’ worldviews, gender, sexual orientation, and cultural orientations. An understanding of this concept enables novice learners to critically examine all evidence—be it a short blog post or a peer-reviewed conference proceeding—and to ask relevant questions about origins, context, and suitability for the current information need. Thus, novice learners come to respect the expertise that authority

represents while remaining skeptical of the systems that have elevated that authority and the information created by it. Experts know how to seek authoritative voices but also recognize that unlikely voices can be authoritative, depending on need. Novice learners may need to rely on basic indicators of authority, such as type of publication or author credentials, where experts recognize schools of thought or discipline-specific paradigms.

Knowledge Practices

Learners who are developing their information literate abilities

- define different types of authority, such as subject expertise (e.g., scholarship), societal position (e.g., public office or title), or special experience (e.g., participating in a historic event);
- use research tools and indicators of authority to determine the credibility of sources, understanding the elements that might temper this credibility;
- understand that many disciplines have acknowledged authorities in the sense of well-known scholars and publications that are widely considered “standard,” and yet, even in those situations, some scholars would challenge the authority of those sources;
- recognize that authoritative content may be packaged formally or informally and may include sources of all media types;
- acknowledge they are developing their own authoritative voices in a particular area and recognize the responsibilities this entails, including seeking accuracy and reliability, respecting intellectual property, and participating in communities of practice;
- understand the increasingly social nature of the information ecosystem where authorities actively connect with one another and sources develop over time.

Dispositions

Learners who are developing their information literate abilities

- develop and maintain an open mind when encountering varied and sometimes conflicting perspectives;
- motivate themselves to find authoritative sources, recognizing that authority may be conferred or manifested in unexpected ways;
- develop awareness of the importance of assessing content with a skeptical stance and with a self-awareness of their own biases and worldview;
- question traditional notions of granting authority and recognize the value of diverse ideas and worldviews;
- are conscious that maintaining these attitudes and actions requires frequent self-evaluation.

APPENDIX B: INFORMATION CREATION AS A PROCESS ***(Framework, pp. 14-15)***

Information in any format is produced to convey a message and is shared via a selected delivery method. The iterative processes of researching, creating, revising, and disseminating information vary, and the resulting product reflects these differences.

The information creation process could result in a range of information formats and modes of delivery, so experts look beyond format when selecting resources to use. The unique capabilities and constraints of each creation process as well as the specific information need determine how the product is used. Experts recognize that information creations are valued differently in different contexts, such as academia or the workplace. Elements that affect or reflect on the creation, such as a pre- or post-publication editing or reviewing process, may be indicators of quality. The dynamic nature of information creation and dissemination requires ongoing attention to understand evolving creation processes. Recognizing the nature of information creation, experts look to the underlying processes of creation as well as the final product to critically evaluate the usefulness of the information. Novice learners begin to recognize the significance of the creation process, leading them to increasingly sophisticated choices when matching information products with their information needs.

Knowledge Practices

Learners who are developing their information literate abilities

- articulate the capabilities and constraints of information developed through various creation processes;
- assess the fit between an information product's creation process and a particular information need;
- articulate the traditional and emerging processes of information creation and dissemination in a particular discipline;
- recognize that information may be perceived differently based on the format in which it is packaged;
- recognize the implications of information formats that contain static or dynamic information;
- monitor the value that is placed upon different types of information products in varying contexts;
- transfer knowledge of capabilities and constraints to new types of information products;
- develop, in their own creation processes, an understanding that their choices impact the purposes for which the information product will be used and the message it conveys.

Dispositions

Learners who are developing their information literate abilities

- are inclined to seek out characteristics of information products that indicate the underlying creation process;
- value the process of matching an information need with an appropriate product;
- accept that the creation of information may begin initially through communicating in a range of formats or modes;
- accept the ambiguity surrounding the potential value of information creation expressed in emerging formats or modes;
- resist the tendency to equate format with the underlying creation process;
- understand that different methods of information dissemination with different purposes are available for their use.

APPENDIX C: INFORMATION HAS VALUE (*Framework*, pp. 16-17)

Information possesses several dimensions of value, including as a commodity, as a means of education, as a means to influence, and as a means of negotiating and understanding the world. Legal and socioeconomic interests influence information production and dissemination.

The value of information is manifested in various contexts, including publishing practices, access to information, the commodification of personal information, and intellectual property laws. The novice learner may struggle to understand the diverse values of information in an environment where “free” information and related services are plentiful and the concept of intellectual property is first encountered through rules of citation or warnings about plagiarism and copyright law. As creators and users of information, experts understand their rights and responsibilities when participating in a community of scholarship. Experts understand that value may be wielded by powerful interests in ways that marginalize certain voices. However, value may also be leveraged by individuals and organizations to effect change and for civic, economic, social, or personal gains. Experts also understand that the individual is responsible for making deliberate and informed choices about when to comply with and when to contest current legal and socioeconomic practices concerning the value of information.

Knowledge Practices

Learners who are developing their information literate abilities

- give credit to the original ideas of others through proper attribution and citation;
- understand that intellectual property is a legal and social construct that varies by culture;
- articulate the purpose and distinguishing characteristics of copyright, fair use, open access, and the public domain;
- understand how and why some individuals or groups of individuals may be underrepresented or systematically marginalized within the systems that produce and disseminate information;
- recognize issues of access or lack of access to information sources;
- decide where and how their information is published;
- understand how the commodification of their personal information and online interactions affects the information they receive and the information they produce or disseminate online;
- make informed choices regarding their online actions in full awareness of issues related to privacy and the commodification of personal information.

Dispositions

Learners who are developing their information literate abilities

- respect the original ideas of others;
- value the skills, time, and effort needed to produce knowledge;
- see themselves as contributors to the information marketplace rather than only consumers of it;
- are inclined to examine their own information privilege.

APPENDIX D: RESEARCH AS INQUIRY (*Framework*, pp. 18-19)

Research is iterative and depends upon asking increasingly complex or new questions whose answers in turn develop additional questions or lines of inquiry in any field.

Experts see inquiry as a process that focuses on problems or questions in a discipline or between disciplines that are open or unresolved. Experts recognize the collaborative effort within a discipline to extend the knowledge in that field. Many times, this process includes points of disagreement where debate and dialogue work to deepen the conversations around knowledge. This process of inquiry extends beyond the academic world to the community at large, and the process of inquiry may focus upon personal, professional, or societal needs. The spectrum of inquiry ranges from asking simple questions that depend upon basic recapitulation of knowledge to increasingly sophisticated abilities to refine research questions, use more advanced research methods, and explore more diverse disciplinary perspectives. Novice learners acquire strategic perspectives on inquiry and a greater repertoire of investigative methods.

Knowledge Practices

Learners who are developing their information literate abilities

- Formulate questions for research based on information gaps or on reexamination of existing, possibly conflicting, information;
- determine an appropriate scope of investigation;
- deal with complex research by breaking complex questions into simple ones, limiting the scope of investigations;
- use various research methods, based on need, circumstance, and type of inquiry;
- monitor gathered information and assess for gaps or weaknesses;
- organize information in meaningful ways;
- synthesize ideas gathered from multiple sources;
- draw reasonable conclusions based on the analysis and interpretation of information.

Dispositions

Learners who are developing their information literate abilities

- consider research as open-ended exploration and engagement with information;
- appreciate that a question may appear to be simple but still disruptive and important to research;
- value intellectual curiosity in developing questions and learning new investigative methods;
- maintain an open mind and a critical stance;
- value persistence, adaptability, and flexibility and recognize that ambiguity can benefit the research process;
- seek multiple perspectives during information gathering and assessment;
- seek appropriate help when needed;
- follow ethical and legal guidelines in gathering and using information;
- demonstrate intellectual humility (i.e., recognize their own intellectual or experiential limitations).

APPENDIX E: SCHOLARSHIP AS CONVERSATION

(Framework, pp. 20-21)

Communities of scholars, researchers, or professionals engage in sustained discourse with new insights and discoveries occurring over time as a result of varied perspectives and interpretations.

Research in scholarly and professional fields is a discursive practice in which ideas are formulated, debated, and weighed against one another over extended periods of time. Instead of seeking discrete answers to complex problems, experts understand that a given issue may be characterized by several competing perspectives as part of an ongoing conversation in which information users and creators come together and negotiate meaning. Experts understand that, while some topics have established answers through this process, a query may not have a single uncontested answer. Experts are therefore inclined to seek out many perspectives, not merely the ones with which they are familiar. These perspectives might be in their own discipline or profession or may be in other fields. While novice learners and experts at all levels can take part in the conversation, established power and authority structures may influence their ability to participate and can privilege certain voices and information. Developing familiarity with the sources of evidence, methods, and modes of discourse in the field assists novice learners to enter the conversation. New forms of scholarly and research conversations provide more avenues in which a wide variety of individuals may have a voice in the conversation. Providing attribution to relevant previous research is also an obligation of participation in the conversation. It enables the conversation to move forward and strengthens one's voice in the conversation.

Knowledge Practices

Learners who are developing their information literate abilities

- cite the contributing work of others in their own information production;
- contribute to scholarly conversation at an appropriate level, such as local online community, guided discussion, undergraduate research journal, conference presentation/poster session;
- identify barriers to entering scholarly conversation via various venues;
- critically evaluate contributions made by others in participatory information environments;
- identify the contribution that particular articles, books, and other scholarly pieces make to disciplinary knowledge;
- summarize the changes in scholarly perspective over time on a particular topic within a specific discipline;
- recognize that a given scholarly work may not represent the only or even the majority perspective on the issue.

Dispositions

Learners who are developing their information literate abilities

- recognize they are often entering into an ongoing scholarly conversation and not a finished conversation;
- seek out conversations taking place in their research area;

- see themselves as contributors to scholarship rather than only consumers of it;
- recognize that scholarly conversations take place in various venues;
- suspend judgment on the value of a particular piece of scholarship until the larger context for the scholarly conversation is better understood;
- understand the responsibility that comes with entering the conversation through participatory channels;
- value user-generated content and evaluate contributions made by others;
- recognize that systems privilege authorities and that not having a fluency in the language and process of a discipline disempowers their ability to participate and engage.

APPENDIX F: SEARCHING AS STRATEGIC EXPLORATION

(Framework, pp. 22-23)

Searching for information is often nonlinear and iterative, requiring the evaluation of a range of information sources and the mental flexibility to pursue alternate avenues as new understanding develops.

The act of searching often begins with a question that directs the act of finding needed information. Encompassing inquiry, discovery, and serendipity, searching identifies both possible relevant sources as well as the means to access those sources. Experts realize that information searching is a contextualized, complex experience that affects, and is affected by, the cognitive, affective, and social dimensions of the searcher. Novice learners may search a limited set of resources, while experts may search more broadly and deeply to determine the most appropriate information within the project scope. Likewise, novice learners tend to use few search strategies, while experts select from various search strategies, depending on the sources, scope, and context of the information need.

Knowledge Practices

Learners who are developing their information literate abilities

- determine the initial scope of the task required to meet their information needs;
- identify interested parties, such as scholars, organizations, governments, and industries, who might produce information about a topic and then determine how to access that information;
- utilize divergent (e.g., brainstorming) and convergent (e.g., selecting the best source) thinking when searching;
- match information needs and search strategies to appropriate search tools;
- design and refine needs and search strategies as necessary, based on search results;
- understand how information systems (i.e., collections of recorded information) are organized in order to access relevant information;
- use different types of searching language (e.g., controlled vocabulary, keywords, natural language) appropriately;
- manage searching processes and results effectively.

Dispositions

Learners who are developing their information literate abilities

- exhibit mental flexibility and creativity;
- understand that first attempts at searching do not always produce adequate results;
- realize that information sources vary greatly in content and format and have varying relevance and value, depending on the needs and nature of the search;
- seek guidance from experts, such as librarians, researchers, and professionals;
- recognize the value of browsing and other serendipitous methods of information gathering;
- persist in the face of search challenges, and know when they have enough information to complete the information task.